

IT3051 - Fundamentals of Data Mining

Dataset Proposal

ReserveIQ

**Machine Learning Based Predictive System for Early Hotel
Booking Cancellation Detection**

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1. Introduction

Hotel cancellations are one of the most disruptive sources of revenue loss for the hospitality industry. A confirmed reservation that is later cancelled often close to the check-in date leaves a room that could otherwise have been sold to another guest, disrupts staffing and inventory planning, and forces hotels to either overbook rooms or accept the empty-room loss. Since online travel agencies and flexible cancellation policies have made cancelling a reservation nearly frictionless for guests, hotels increasingly need a way to anticipate which bookings are likely to be honored versus cancelled, well before the arrival date.

This project proposes a machine learning system to predict, at the time a reservation exists in the system, whether it is likely to end in cancellation. Instead of treating every booking with the same level of operational planning, the hotel could flag high-risk reservations early and act on them for example through targeted confirmation reminders, deposit requirements, or smarter overbooking decisions.

2. Problem Statement

Given the attributes of an existing hotel reservation, including how and when it was booked, the details of the planned stay, the type of customer making the booking, the price agreed, and that guest's prior booking history with the hotel, can a machine learning model reliably predict whether the reservation will be cancelled before the stay occurs?

This is a supervised binary classification problem: each reservation is labelled as either “cancelled” or “not cancelled” (honoured/checked-out), using only information that is known at or near booking time not information that only becomes available afterward, such as the final reservation status close to or after the arrival date, which would leak the answer.

3. Aim and Objectives

Aim: To design, build, and evaluate a machine learning model that predicts the likelihood that a given hotel reservation will be cancelled, and to present that prediction through an accessible interface that hotel staff could realistically use.

Objectives

- Acquire and understand a realistic hotel-reservations dataset with a clearly defined cancellation label.
- Perform exploratory data analysis to understand booking patterns, class balance between cancelled and honoured reservations, and relationships between features such as lead time, price, and cancellation rate.
- Identify and justify which features are legitimately available at the time a cancellation-risk prediction would be made, avoiding leakage from fields that are only populated after the outcome is known.
- Train and compare at least four classification models (e.g., Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting/XGBoost) for cancellation prediction.
- Evaluate the models using metrics suited to a classification task with class imbalance, rather than relying on accuracy alone.
- Present predictions through a simple interactive interface (e.g., a Streamlit app) where a staff member could enter reservation details and receive a cancellation-risk score.
- Discuss the practical limitations, ethical considerations, and business implications of deploying such a model in a real hotel.

4. Dataset

Dataset: Hotel Booking Demand dataset, based on the data published by Antonio, de Almeida, and Nunes (2019) in *Data in Brief*. The version used in this project is provided as a single CSV file containing 119,390 reservation records, combining bookings from two real Portuguese hotels, a resort hotel (Algarve) and a city hotel (Lisbon) — distinguished by a hotel column. All records cover bookings due to arrive between 1 July 2015 and 31 August 2017. All hotel- and guest-identifying information was removed by the original authors before release.

The variables map directly onto the five dimensions in the problem statement:

Category	Example columns
Booking behaviour	lead_time, booking_changes, deposit_type, distribution_channel, market_segment, days_in_waiting_list
Stay details	stays_in_weekend_nights, stays_in_week_nights, adults, children, babies, meal, reserved_room_type, assigned_room_type
Customer type	customer_type (Transient, Contract, Group, Transient-Party), country, is_repeated_guest
Pricing	adr (average daily rate)
Reservation history	previous_cancellations, previous_bookings_not_canceled
Target	is_canceled (1 = cancelled, 0 = not cancelled)

A key strength of this dataset, noted by the authors themselves, is that it was purpose-built to avoid leakage: rather than pulling variable values straight from the live bookings table, the authors extracted them from the bookings change log, timestamped to the day before arrival so (aside from the two fields excluded below) the feature values reflect what was actually known at that point, not information that only became accurate later (e.g., final guest count, true nationality confirmed at check-in).

Dataset reference: Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. *Data in Brief*, 22, 41-49. <https://doi.org/10.1016/j.dib.2018.11.126> (CC BY 4.0)

Dataset link : <https://www.kaggle.com/datasets/mojtaba142/hotel-booking>

5. Machine Learning Task

This is a supervised binary classification task. Each reservation record is mapped to one of two classes: NOT CANCELLED (0) or CANCELLED (1), using booking behavior, stay details, customer type, pricing, and reservation-history features as input. The model outputs a predicted class and a probability score representing cancellation risk, which can be used to rank reservations by how much attention they warrant.

6. Proposed Methodology

6.1 Data Understanding and Preparation

- Note that NULL in Agent and Company is not missing data the authors state it means “not applicable” (e.g., no travel agent was involved), so it should be encoded as its own category rather than imputed.
- Because the dataset combines two different hotel types (resort and city) under the hotel column, cancellation rates and patterns should be compared across the two categories during EDA rather than assumed to be uniform.
- The authors flag that distributions of some fields (Adults, Children, Meal, Country, AssignedRoomType) genuinely differ between cancelled and non-cancelled bookings, partly because hotels often only finalize details like nationality or guest count at check-in. This is expected and worth discussing rather than treated as a data quality issue.

6.2 Data Preprocessing

- **Missing values**
The dataset contains missing values in four columns: company (112,593 missing), agent (16,340 missing), country (488 missing), and children (4 missing). company and agent will have missing values encoded as an explicit “Not Applicable” category rather than imputed, since a NULL in these fields represents no company/agent being involved in the booking, not a data-entry gap. country will be imputed with the mode or an “Unknown” category, and the 4 missing children values will be imputed with 0 or the median.
- **Duplicate checking**
No exact duplicate rows were found in the dataset; this check will still be explicitly documented as part of the data-cleaning process to demonstrate due diligence.
- **Target leakage**
reservation_status and reservation_status_date will be removed, as both directly encode or timestamp the final outcome and would leak the target into the features.

- **Irrelevant / privacy fields**
name, email, phone-number, and credit_card will be removed. These are personally identifiable fields with no legitimate predictive value for cancellation risk, and retaining them would raise privacy and ethical concerns.
- **Categorical encoding**
Categorical fields: hotel, meal, country, market_segment, distribution_channel, reserved_room_type, assigned_room_type, deposit_type, customer_type will be encoded appropriately (one-hot encoding for low-cardinality fields; frequency or target encoding for high-cardinality fields such as country).
- **Numerical scaling**
Numeric features will be scaled (e.g., standardization) for models sensitive to feature magnitude, Logistic Regression, SVM, and KNN, if used, while tree-based models (Decision Tree, Random Forest, Gradient Boosting) will use unscaled inputs.
- **Outlier handling**
Outliers will be investigated in adr, lead_time, guest counts (adults, children, babies), days_in_waiting_list, and stay length (stays_in_weekend_nights, stays_in_week_nights), using summary statistics and visualizations to identify and handle extreme or implausible values (e.g., negative or unusually high ADR).
- **Feature engineering**
New features will be derived to strengthen predictive signal, including: total_stay (weekend nights + week nights), total_guests (adults + children + babies), a room-change indicator (whether assigned_room_type differs from reserved_room_type), a previous-cancellation flag (derived from previous_cancellations), and a season feature derived from the arrival month.
- **Feature selection**
Weak, redundant, high-cardinality-without-value, privacy-related, and leakage-prone variables will be removed or carefully justified before finalizing the feature set used for modelling.
- **Class balance**
The target class distribution is approximately 37% cancelled vs. 63% not cancelled. While not extreme, this imbalance is still non-trivial and will be assessed during EDA to decide whether class weighting or resampling (e.g., SMOTE) is warranted, and evaluation will use imbalance-aware metrics regardless (Section 6.6).
- **Train/test split**
A stratified train/test split will be used to preserve the cancelled/not-cancelled ratio across both sets, with care taken to fit any preprocessing steps (encoders, scalers, imputers) only on the training set to avoid leakage into the test set.
- **High-cardinality features**
country, agent, and company require careful treatment due to their large number of categories, handled through frequency/target encoding or grouping rare categories into an

“Other” bucket, rather than naive one-hot encoding, which would create an unmanageable number of columns.

6.3 Feature Selection

Candidate predictive features fall into the five categories the problem statement specifies: booking behaviour (lead time, deposit type, market segment, distribution channel, booking changes), stay details (length of stay, room type, meal plan, number of guests), customer type (transient/contract/group, repeat-guest status), pricing (average daily rate), and reservation history (prior cancellations, prior completed bookings). Identifier-like or administrative fields with limited predictive value (e.g., reservation_status_date) will be excluded or reserved only for leakage checks.

6.4 Data Leakage Prevention

Two fields must be excluded from the feature set entirely:

- ReservationStatus - a three-way restatement of the outcome (Canceled / Check-Out / No-Show), directly derivable from the target.
- ReservationStatusDate - the date that status was set, only known after the outcome.

AssignedRoomType needs care: it can differ from ReservedRoomType due to hotel-side operational decisions (e.g. overbooking reassignment) that may happen close to or after the cancellation decision -this will be checked during EDA before deciding whether to keep, drop, or replace it with a derived “room type changed” flag.

6.5 Model Development

- Establish a baseline using Logistic Regression.
- Train and compare at least four classification models: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting (XGBoost/LightGBM).
- Encode categorical fields (customer type, market segment, deposit type, meal, country) appropriately, and scale numerical features where the model requires it.
- Address any class imbalance between cancelled and non-cancelled bookings using class weighting or resampling methods (e.g., SMOTE), if the imbalance proves significant.
- Tune hyperparameters using cross-validation on the training set.

6.6 Evaluation

Since cancellations, while common in this dataset, are not necessarily a small minority class (historical figures for this dataset show a meaningful cancellation rate), the exact class balance will be measured first and the evaluation approach adjusted accordingly. Models will be compared using Precision, Recall, F1-score, ROC-AUC, PR-AUC, and a confusion matrix on a held-out test set, with particular attention to recall on the cancellation class, since missing a likely cancellation is more costly to the hotel than a false alarm.

6.7 Deployment / Presentation

The selected model and its preprocessing pipeline will be wrapped into an end-to-end prediction flow: a user (e.g., front-desk or revenue-management staff) enters reservation details, the same preprocessing is applied, and a cancellation-risk score is returned. This will be demonstrated through a lightweight Streamlit or web-based interface.

6.8 Potential Users and Decision Support

Potential users include hotel revenue managers, front-desk staff, and reservations teams. The model is intended to support decisions such as which bookings to follow up on with a confirmation call or email, which reservations might justify a deposit requirement, and how aggressively to overbook given the current mix of high- and low-risk reservations. The prediction is decision support only it does not automatically cancel, modify, or overbook any reservation.

7. Tools and Technologies

Category	Tools
Language	Python
Data Handling	Pandas, NumPy
Modelling	Scikit-learn, XGBoost / LightGBM
Visualisation	Matplotlib, Seaborn
Interface	Streamlit / Web
Environment	Jupyter Notebook

8. Expected Outcomes

- A cleaned, well-understood hotel reservations dataset with cancellation as the labelled outcome.
- A trained and validated binary classification model predicting cancellation risk from booking, stay, customer, pricing, and history features.
- A comparative evaluation of at least four classification algorithms using metrics appropriate to the observed class balance.
- A working prototype interface producing a cancellation-risk score for a given reservation.
- A discussion of the model's practical value, limitations, and ethical considerations for hotel operations.

9. Limitations and Ethical Considerations

- Geographic and temporal scope: data comes from two Portuguese hotels over 2015-2017; cancellation behaviour and its drivers may differ in other markets or post-pandemic booking patterns.
- Missing data: agent and company fields have substantial missingness and require a clearly justified imputation or exclusion strategy.
- Fairness considerations: features such as country or customer_type could act as proxies for nationality or group characteristics; the model's use should be limited to operational risk scoring, not justify differential treatment of guests.
- Decision-support framing: predictions should inform staff follow-up and planning, not automatically penalise guests (e.g., automatic deposit charges or cancellations) without human review.
- NULL vs. missing: Agent/Company NULLs are a legitimate “not applicable” category per the source paper, not missing data, mishandling this (e.g., dropping rows) would needlessly shrink the dataset.
- Two distinct properties in one dataset: the data spans a resort hotel and a city hotel with potentially different cancellation drivers; the model should be validated on each hotel subgroup separately to ensure it isn't just learning hotel-specific quirks